Imperial College London

Department of Earth Science and Engineering

MSc in Environmental Data Science and Machine Learning

Independent Research Project

Project Plan

From Pixels to Point Clouds: GroundDINO-Driven 2D-3D Segmentation for High-Resolution Carbon Stock Assessment

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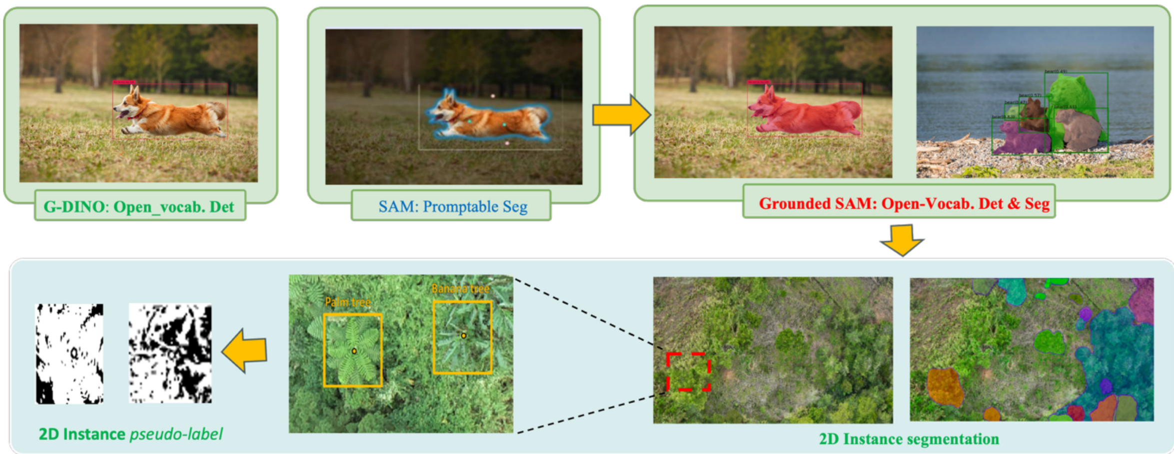

Figure 1. Grounded SAM usage in 2D UAV images instance segmentation

Abstract

Accurate monoculture biomass and carbon-stock estimates are essential for nature-based economic restoration of abandoned mine sites, yet field surveys and fully supervised 3D deep-learning approaches impose prohibitive costs. This study introduces a streamlined, low-labeling workflow that fuses centimeters-resolution UAV multispectral orthophotos with co-registered LiDAR point clouds to generate a digital-twin carbon map of a decommissioned copper mine in Tublao, Baguio Province, Philippines. **Open-vocabulary detection (Grounded DINO)** first locates individual tree crowns, which are then refined by **SAM** into pixel-accurate masks. Those 2D masks are geometrically projected onto the point cloud to produce 3D instance pseudo-labels, guiding a self-distillation-enhanced **Segment3D** network to perform single-instance segmentation without large-scale annotation. Extracted metrics—tree height and crown spread—are converted into above-ground biomass using species-specific allometric equations and the IPCC 0.47 carbon factor, with uncertainty quantified via Monte Carlo simulation. The resulting workflow dramatically reduces annotation costs, delivers high-resolution carbon estimates, and compellingly demonstrates the transformative potential of foundation models in ecological mine-closure planning.

1 Introduction

Rapid growth in demand for low-carbon energy and key minerals means that mining will reach record levels over the next 50 years, yet poorly managed mine closures put ecosystems and communities at risk. Due to insufficient ecological restoration, many abandoned or decommissioned mine sites remain degraded for extended periods, posing a continuous threat to soil, surface water, and community safety, and severely hindering the social license and sustainability of future mining activities[1].

To address these challenges, nature-based restoration strategies and natural capital accounting are increasingly becoming central to mine restoration planning[1]. In this context, high-resolution unmanned aerial remote sensing - especially multispectral imagery with LiDAR scanning - offers unprecedented possibilities for reconstructing post-closure landscapes of mines with monoculture-level accuracy[2]. However, there are still technical bottlenecks in extracting actionable ecological metrics such as species diversity, biomass, or carbon stock from these data: on the one hand, traditional semantic segmentation methods can only differentiate between “vegetation/non-vegetation” and lack of instance-level information; on the other hand, conventional point cloud instance segmentation is highly dependent on large-scale manual labelling, which makes it difficult to maintain the accuracy in complex forest stand environments[3].

Recently, advances in open-domain visual models and 3D infrastructure networks have opened up new paths for automatic monoculture segmentation without large-scale labelling: Grounded DINO and Segment Anything (SAM)[4] are able to achieve instance-aware segmentation based on linguistic cues on orthophotos, while Segment3D accomplishes instance-independent segmentation by projecting 2D pseudo-labels onto the point cloud and combining with self-supervised distillation to accomplish category-independent 3D instance segmentation.[3] Although each of these methods has been validated in generalized scenarios, few studies have integrated them into a complete workflow for ecological restoration and carbon monitoring.

In this study, we selected a site at the abandoned Santo Niño copper mine in Tublao, Baguio Province, Philippines,[5] with the help of high-resolution multispectral orthophotos (GSD (Ground Sampling Distance) $\approx$ 2-3 cm) and high-density LiDAR point clouds ($\geqslant$ 300 pts/m ^{2}) provided by the Bio+Mine project[1], and have completed a community participatory baseline survey and an administrative licensing process. The pipeline is a multi-stage process(section 2.1 to section 2.6). This workflow not only provides a refined natural capital assessment tool for the ecological restoration of abandoned mines but also demonstrates the practical value of Grounded DINO SAM[4] and Segment3D[3] in ecological remote sensing.

2 Literature Review

UAV Remote Sensing and Carbon Stock Estimation. UAV multispectral imagery and LiDAR point clouds have been widely used for high-precision estimation of biomass and carbon stocks in forest stands. Ayrey and Hayes[6] in 2018 systematically evaluated a variety of LiDAR-based algorithms for single-tree segmentation and reported a performance of 0.65-

0.80 for mIoU in different forest stands. Räsänen et al.[7] used UAV multispectral imagery and LiDAR to model forest carbon density and achieved <20 % estimation error. These works laid the foundation for the reliability of UAV remote sensing in ecological assessment.

2D Instance Segmentation Methods. Traditional 2D semantic segmentation[8](e.g., UNet, DeepLab) has difficulty distinguishing canopy instances that are occluded from each other. The latest open-domain detection and segmentation method, Grounded DINO[9], combined with natural language cues, can detect canopy locations with zero samples on orthophotos, while SAM can refine the detection frame into a high-resolution pixel mask with an IoU improvement of about 8 %[10].

3D Instance Segmentation with Self-Supervision. Point cloud instance segmentation often relies on manual labelling or rule-based clustering (e.g., region growing), but it is difficult to generalize in heterogeneous canopy environments. Segment3D[3] proposes to utilize 2D pseudo-labels to do 3D pre-training, and then refine the model by self-supervised distillation, which not only saves labelling cost but also achieves mIoU > 0.75 in multiple scenes[11].

Cross-modal pseudo-label projection. The feasibility of 2D→3D projection for pseudo-label generation has been demonstrated on indoor RGB-D datasets. In an outdoor ecological scene, Xie et al.[12] used the Digital Surface Model (DSM) / Canopy Height Model (CHM) as a "depth channel" to map the mask to the point cloud at the pixel level, and the mIoU was improved by about 5-10 %. This shows that high-precision projection is the key to realize cross-modal segmentation.

2 Methodology

This section describes the full multi-stage workflow—from raw UAV products to tree-level carbon estimates. Fig1 shows that Grounded SAM can segment the 2D UAV images and generate 2D pseudo-labels[4]. After 2D images and labels have been projected to 3D space, Segment3D can achieve 3D point cloud segmentation. The Segment3D Framework is displayed in Fig2.

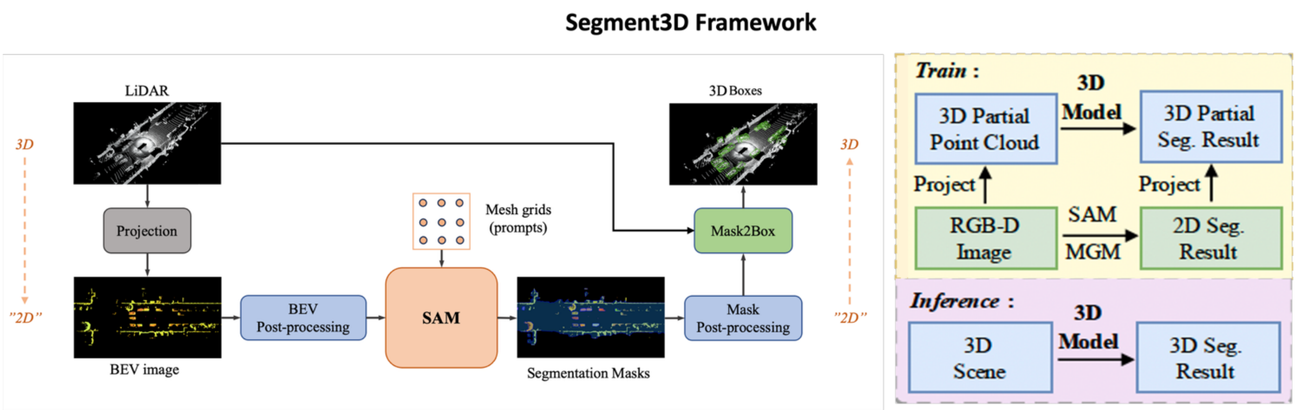

Figure 2 Segment3D framework

2.1 Data acquisition and co-registration

Orthorectified multispectral mosaics and high-density UAV-LiDAR point clouds were acquired over an abandoned copper mine in Tublao, Philippines, by the Bio+Mine consortium[1]. Both data sets are re-projected to **WGS-84 / UTM 51 N (EPSG:32651)**. A digital surface model and canopy-height model (CHM=DSM-DTM) are generated with **PDAL**. Residual planar offsets (<0.2m RMSE) are removed by point-to-mesh ICP in **CloudCompare**.

2.2 2-D instance proposal and refinement

Open-vocabulary detector **Grounding DINO** is prompted with a "tree crown" to produce bounding boxes on each Ortho mosaic tile[12]. Boxes with (confidence<0.35) are discarded. Each remaining box is supplied to **Segment Anything (SAM)**, yielding a **high-resolution** mask. Masks are refined by:

(1) Sharpening edges using dense conditional random fields (CRF).
(2) 3×3 morphological close/open operations.
(3) region size and confidence-based filtering (regions $\geq$ 200 pixels, mean score $\geq$ 0.50).
(4) This hybrid strategy improves the mIoU of the zero-sample SAM mask by about 8 under closed canopy conditions[10].

2.3 2-D to 3-D mask projection

Each refined mask is back-projected onto the LiDAR cloud:

(1) Pixel centers (u, v) → world (X, Y) via the orthomosaic affine matrix.
(2) All points whose horizontal distance to (X, Y) is $<$ ½ GSD and whose height Z is within $\pm 1\,m$ of CHM is selected.
(3) Selected points inherit the mask's instance ID. Overlapping views are fused by majority voting.
(4) The output is a set of point-cloud tiles with **instance-level** pseudo-labels $\mathcal{L}_{\boldsymbol{pseudo}}$.

2.4 Segment3D: 3D instance segmentation

Segment3D is a Transformer-based network trained in two stages[3]:
Stage 1 - Pseudo-label pre-training. 10 m × 10 m × 15 m tiles containing ⩾2 labelled canopies are vowelized (with a resolution of 5 cm) and the network is trained with the generated pseudo-label supervision.
Stage 2 - Self-supervised distillation. Inference of the full-field attraction cloud using the pre-trained model Predictions with SoftMax confidence $>$ 0.9 are added to the pseudo-label set and the model is fine-tuned for five epochs to reduce label noise.

2.5 Tree-level feature extraction and carbon estimation

For each segmented tree $\boldsymbol{i}$:

Table 1: Tree-level geometric features extracted from the segmented point cloud

Symbol	Feature	Computation
H_i	height	$\boldsymbol{max}(Z) - Z_{ground}$
$A_{c,i}$	crown area	area of 2-D convex hull
W_i	crown width	mean of hull major/minor axes
D_i	DBH	$f_{GBM}(H_i, A_{c,i}, W_i)$ (cross-validated LightGBM)

Above-ground biomass (AGB) is estimated with a moist-tropical allometry[13]:

$$AGB_i = \rho e^{(a+blnDi+clnHi)} \tag{1}$$

where ρ is wood density (kg/m^{-3}) and $\boldsymbol{a, b, c}$ are species-level coefficients. Carbon content is

$$C_i = 0.47 \times AGB_i \tag{2}$$

Uncertainty is quantified via a 1 000-run Monte-Carlo simulation perturbing H_i , D_i, ρ, $(\mathrm{a, b, c})$ within reported standard error; $95\ \%$ confidence intervals are given for total site-level carbon $C = \sum_i C_i$.

2.6 Limitations and Mitigation

1. **Data alignment error:** residual error > 20 cm → automatically apply local affine correction.
2. **Shaded area mask failure:** if shaded area 2D mIoU < 0.70 → use 100 manually labeled tiles to fine-tune SAMs.
3. **GPU resource bottleneck:** cluster queue > 24 hours → backup call to Azure Spot GPU (budget 72 hours).
4. **LiDAR point cloud voids:** missing point cloud areas > 5 % → fill with SfM-generated point clouds.

3 Evaluation plan

To verify that the proposed workflow meets the accuracy required for tree-level carbon accounting, three complementary metrics will be reported on a hand-labelled validation subset (~500 crowns):

Table 2: Evaluation Plan

Metric	Rationale	Target
mIoU(Eq.3)	Overall volumetric overlap between predicted and reference crowns	≥ 0.75
Boundary F-score (BF)(Eq.4)	Sensitivity to over/under-segmentation at the crown edge	≥ 0.80
Carbon RMSE[14]	Accuracy of carbon-stock estimation versus field plots	≤ 15 %

Mean Intersection-over-Union (mIoU). For each tree instance k

$$IoU_k = \frac{|P_k \cap G_k|}{|P_k \cup G_k|} \tag{3}$$

where P_k and G_k are predicted and ground-truth point sets. mIoU is the mean over all k instances[15].
Boundary F-score (BF). Boundaries of $P_k$and G_k are dilated by $\delta = 2$ px ($\approx 6\,cm$). Precision P is the fraction of predicted boundary points falling inside the dilated ground-truth boundary; recall R is the converse.

$$BF = 2PR/(P + R) \tag{4}$$

BF penalizes over-/under-segmentation near crown edges[16].

4 Expected Deliverables

1. **Open-source pipeline**. Docker-based implementation of Ground DINO → SAM → Segment3D→Carbon Stock Map.
2. **High-resolution products.** Tree-instance point-cloud layer (LAZ + GeoJSON)10 cm biomass & carbon raster's (COG / GeoTIFF)
3. **Technical reports.** mid-term progress memo, final 5,000-word thesis formatted as a journal article.
4. **Interactive dashboard**. Web-GL viewer for crown segmentation and carbon layers, facilitating stakeholder engagement.

To ensure reproducibility and accessibility all results will be saved to One drive and GitHub personal IRP repositories.

5 Future Plan

The Figure3 Gantt chart below outlines the plan for the whole project. One will notice that the work currently done is extensive, yet intangible.

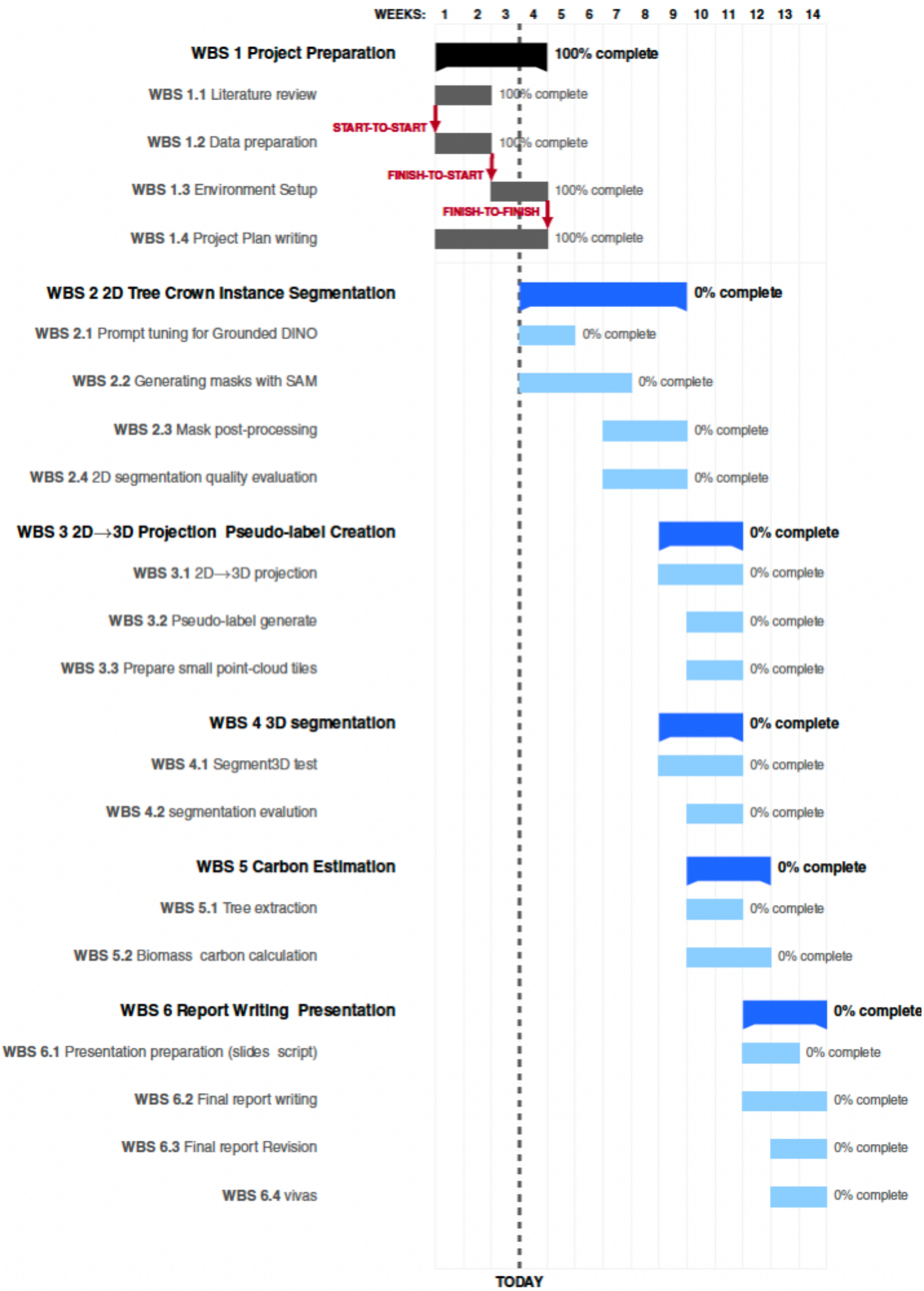

Figure 3: Gantt chart for the project plan

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